Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.21981594217848488, median 0.2016832510144972, std: 0.12763288042828927
Gyroscope error (imu0): mean 1.4602958890709512, median 0.9844449257473694, std: 1.457620677081513
Accelerometer error (imu0): mean 5.779227426774233, median 3.4956676790817984, std: 6.202893500113515

Residuals

Reprojection error (cam0) [px]: mean 0.21981594217848488, median 0.2016832510144972, std: 0.12763288042828927
Gyroscope error (imu0) [rad/s]: mean 0.0021532816800594297, median 0.0014516148675785818, std: 0.002149336941866133
Accelerometer error (imu0) [m/s^2]: mean 0.01860880939236773, median 0.01125586669902044, std: 0.01997299194179616

Transformation (cam0):

T_ci: (imu0 to cam0):
[[-0.00735841 0.99995878 0.00531939 -0.00614862]
 [0.00733555 0.00537337 -0.99995866 0.01569493]
 [-0.99994602 -0.00731909 -0.00737479 -0.18951634]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[-0.00735841 0.00733555 -0.99994602 -0.18966649]
 [0.99995878 0.00537337 -0.00731909 0.00467695]
 [0.00531939 -0.99995866 -0.00737479 0.01432935]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.005712188749342302

Gravity vector in target coords: [m/s^2]
[0.19504141 -9.48062142 -2.49963981]

Calibration configuration
==========

cam0

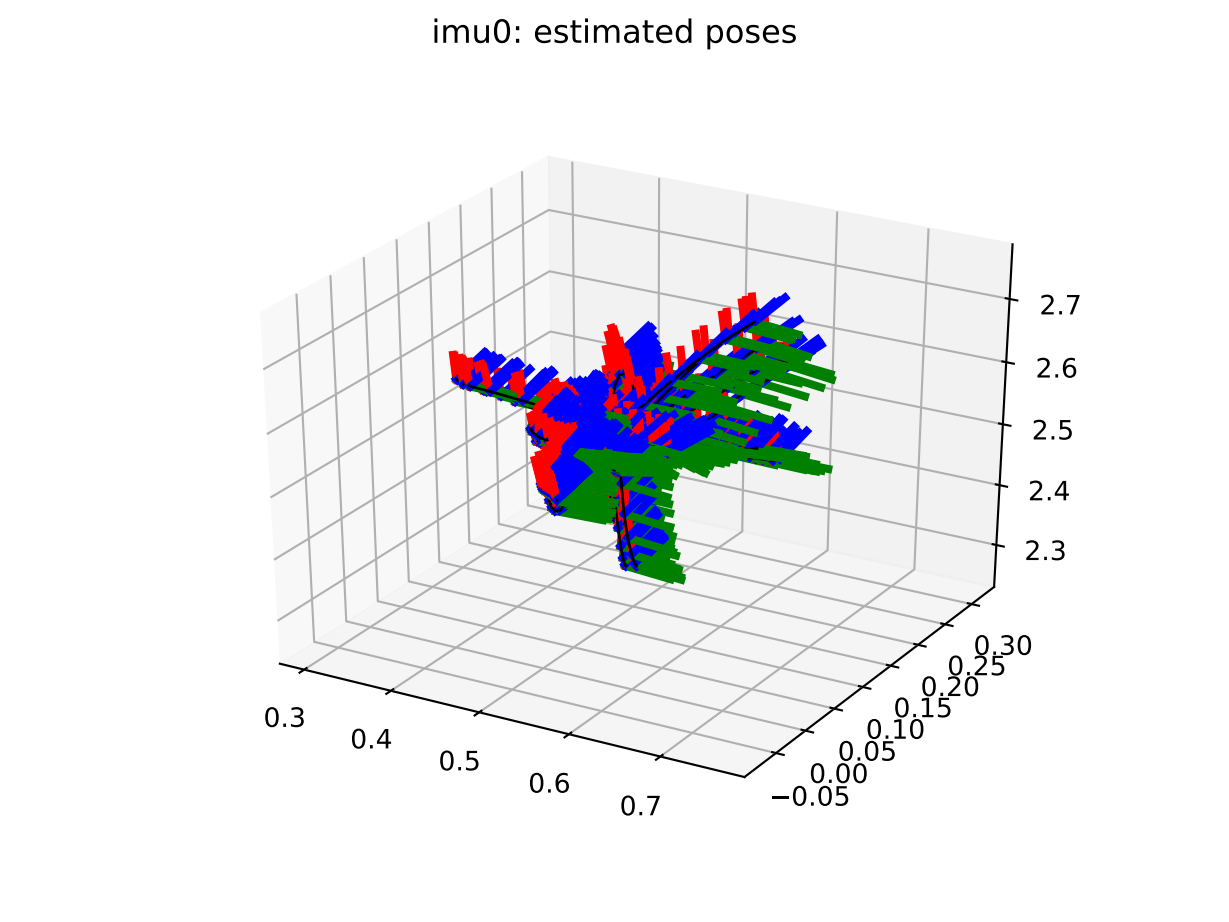
Camera model: pinhole
Focal length: [1006.0201996315349, 1007.7691645918724]
Principal point: [707.1212739368813, 544.5528210405221]
Distortion model: radtan
Distortion coefficients: [-0.39444064173362137, 0.17985588555745766, 0.0004144828653939424 7, -0.00020058863982859635]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.07 [m]
Spacing 0.021 [m]

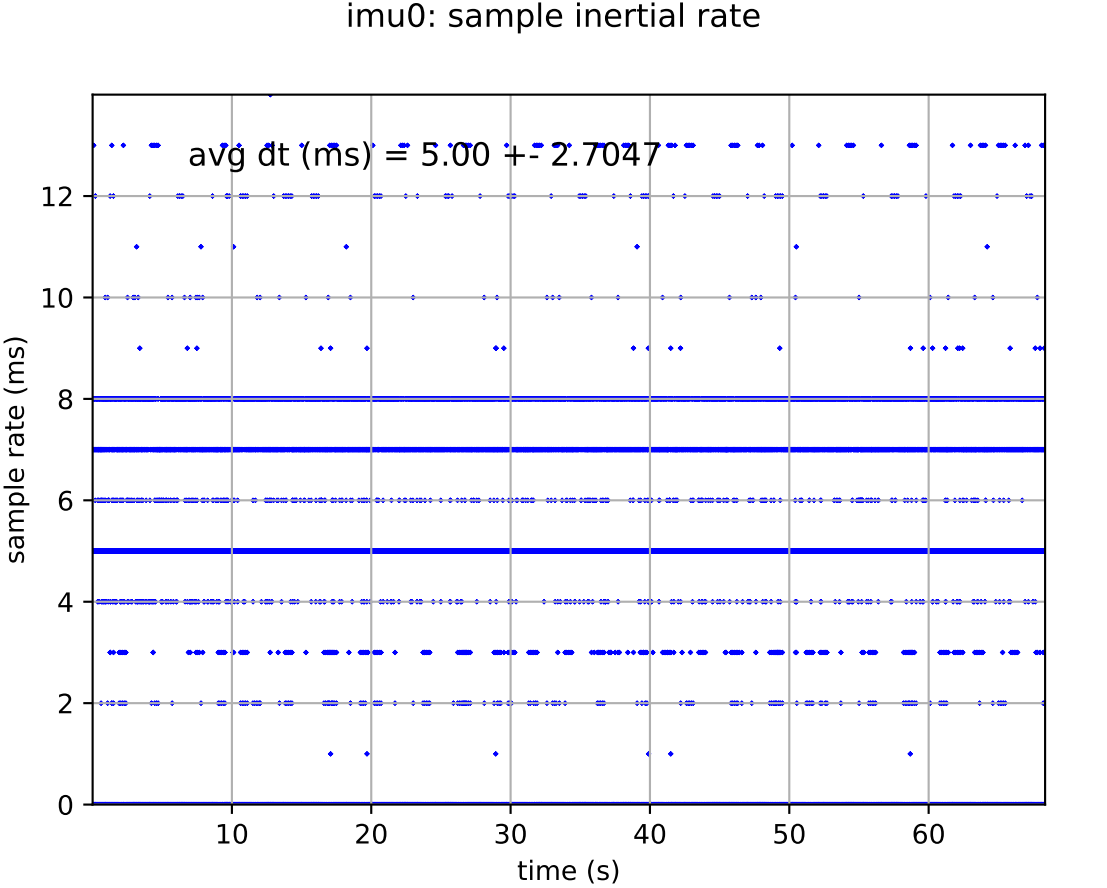
IMU configuration
==========

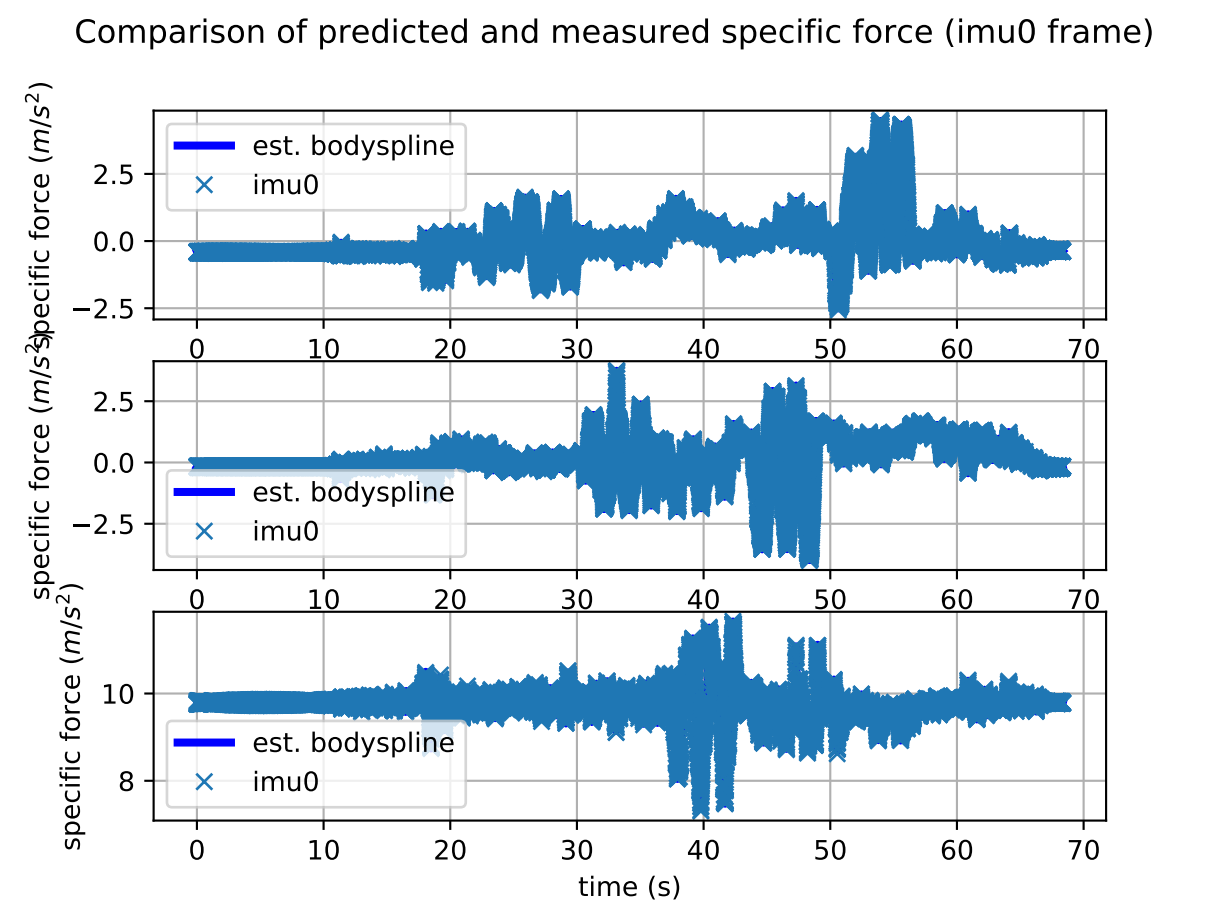
IMU0:

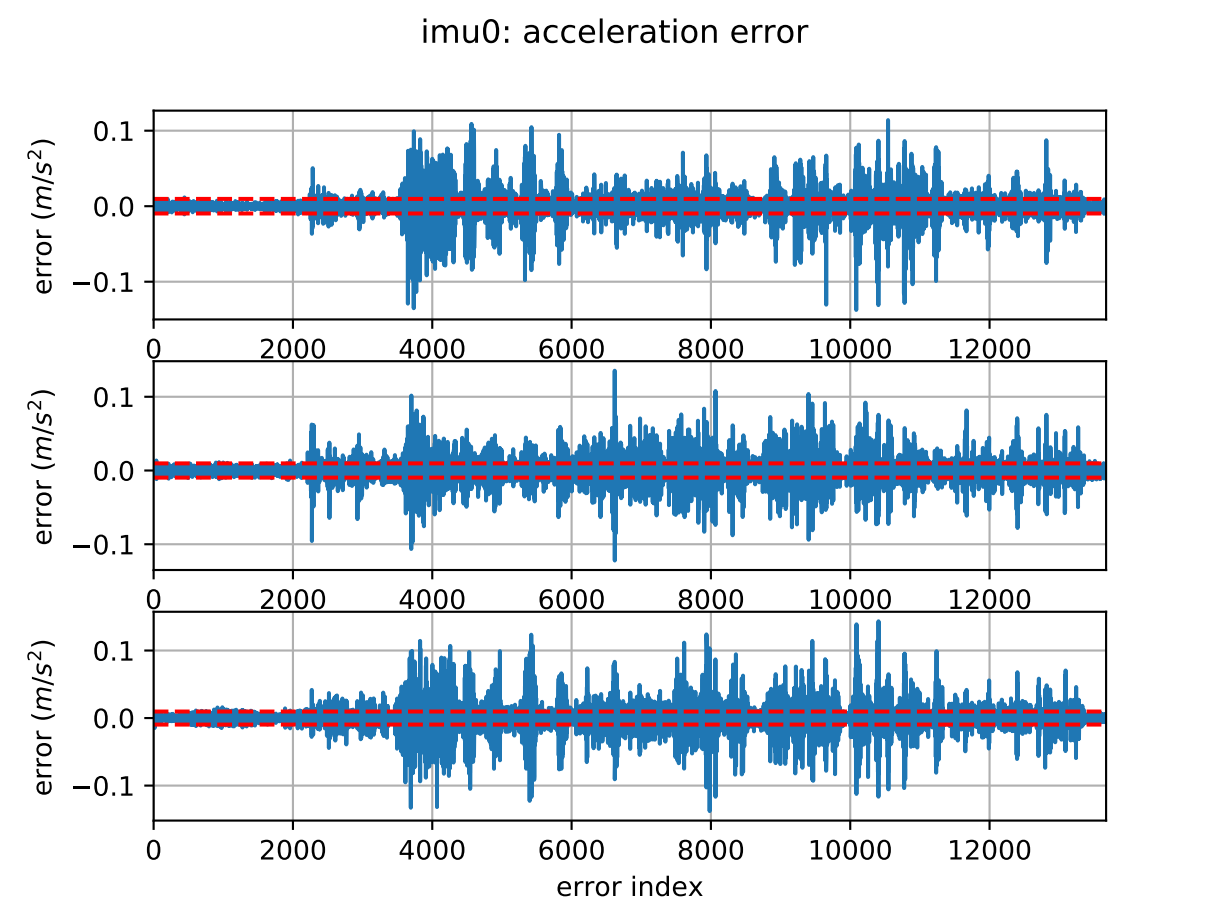
Model: calibrated
Update rate: 200.0
Accelerometer:
Noise density: 0.0002276846771973415
Noise density (discrete): 0.003219947584370 2054
Random walk: 6.706087584689249e-06
Gyroscope:
Noise density: 0.00010426654551109309
Noise density (discrete): 0.001474551627635794
Random walk: 3.5636559575381104e-06

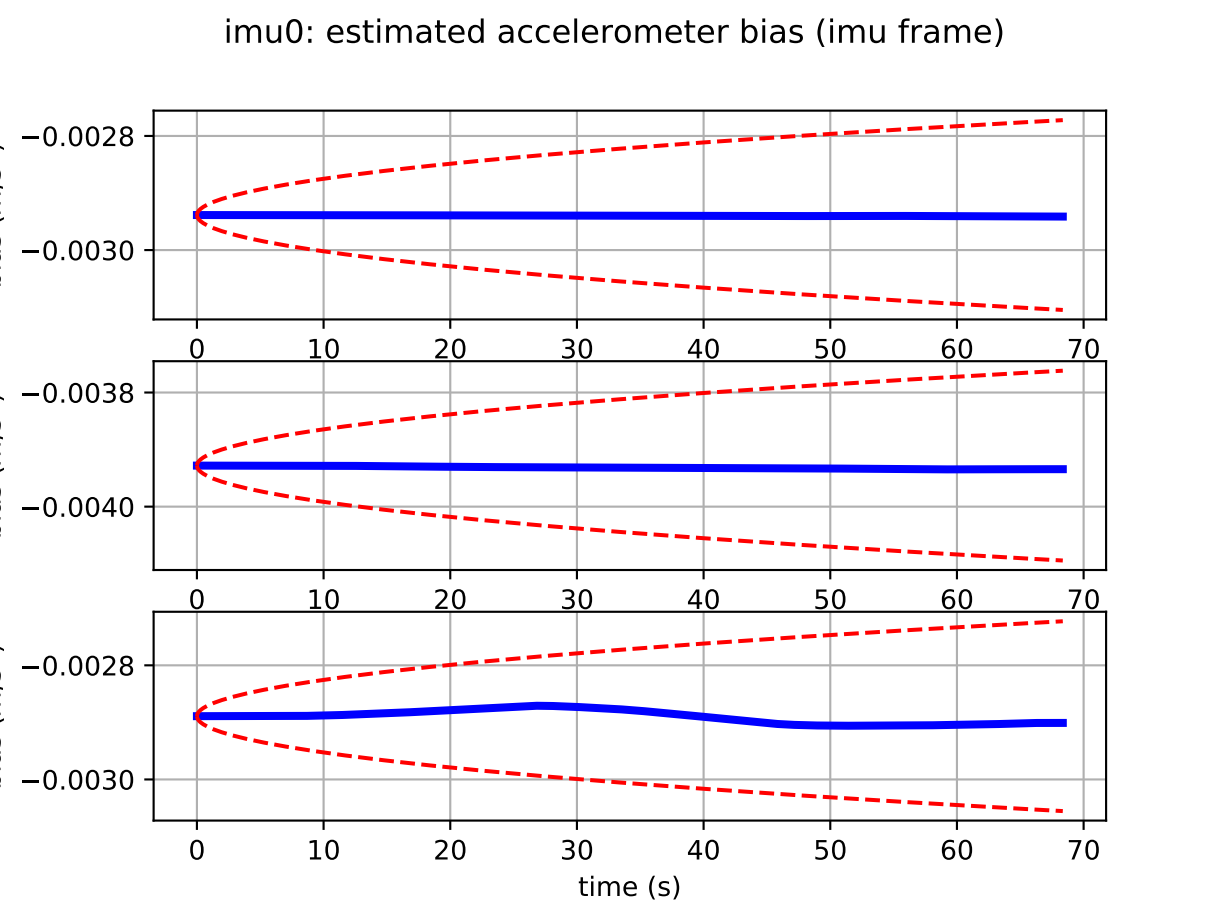
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

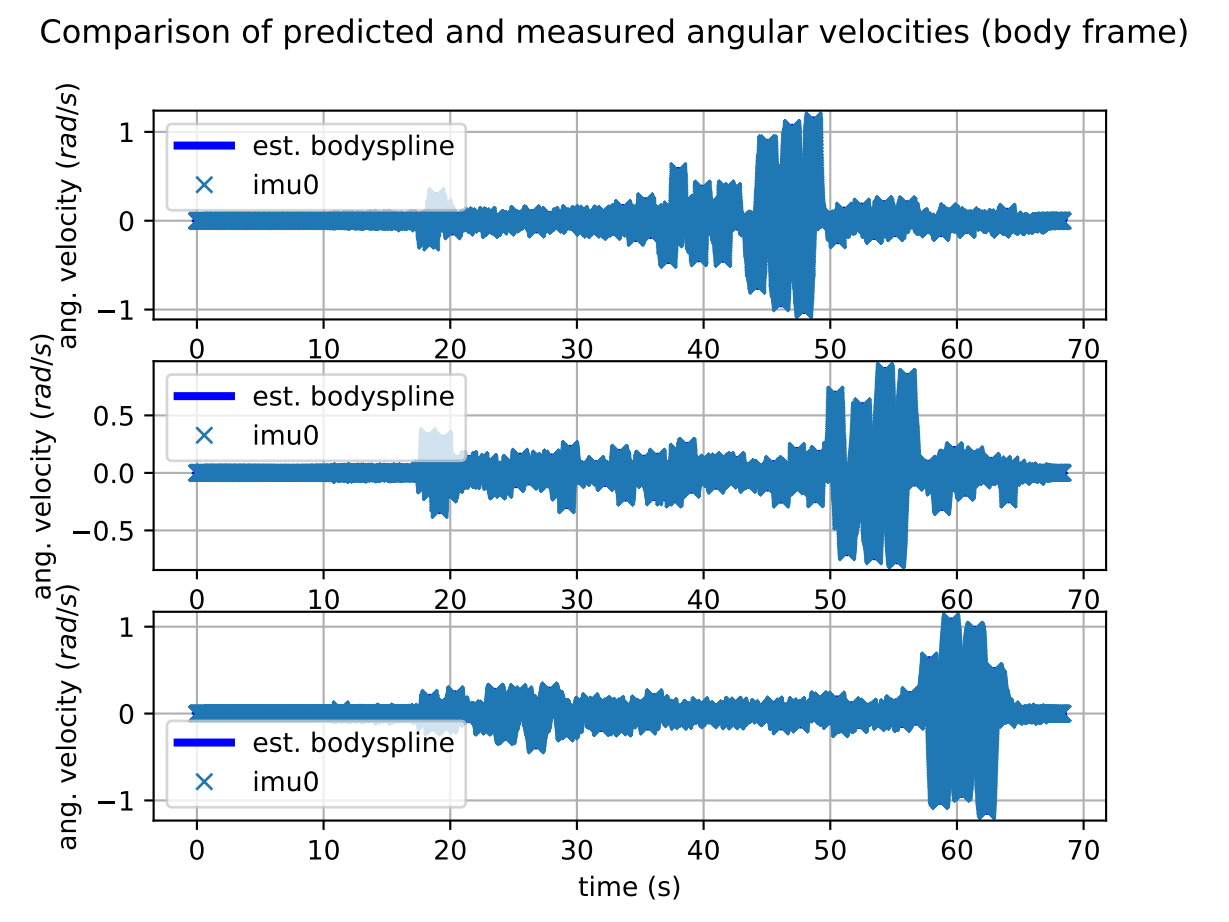

imu0: estimated poses
2.7
2.6
2.5
2.4
2.3
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
0.3
0.4
0.5
0.6
0.7

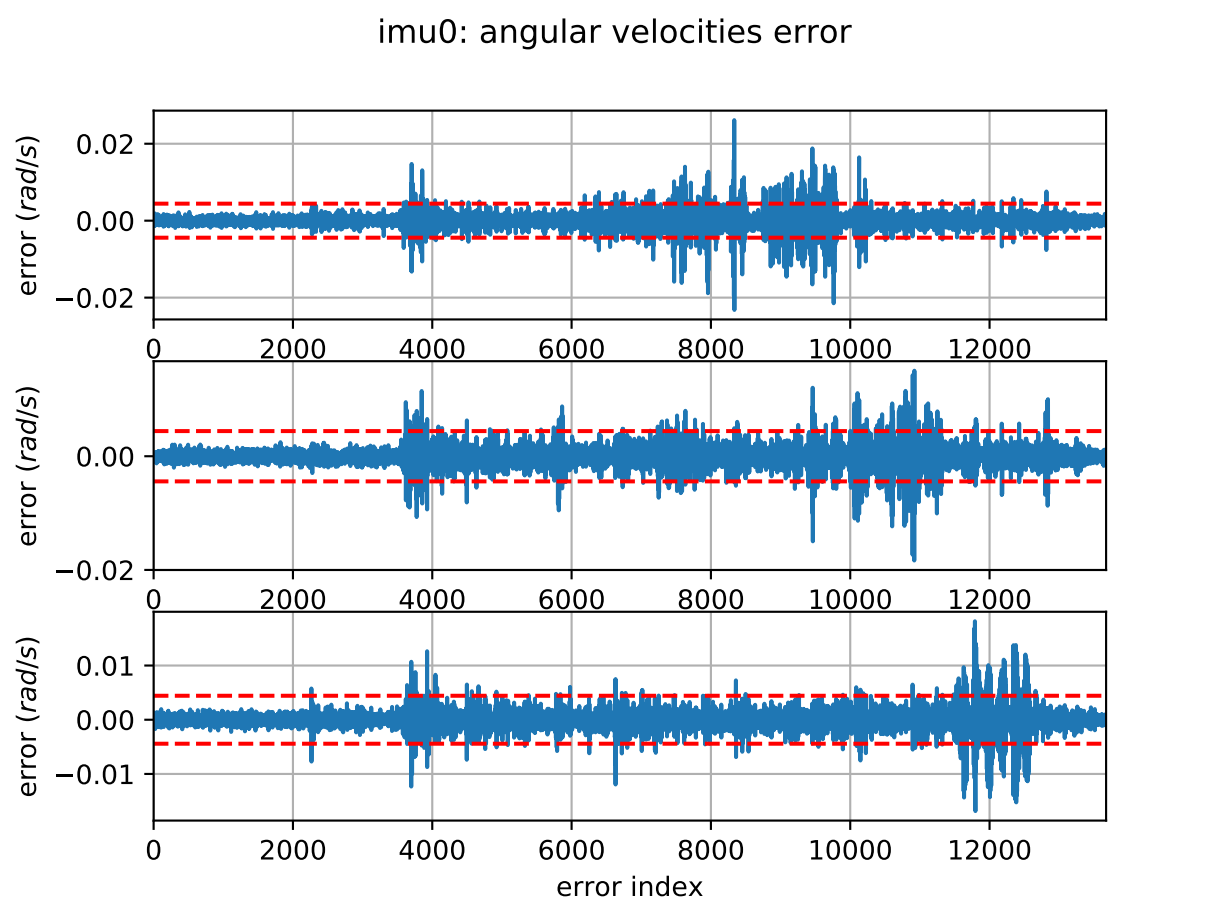

imu0: sample inertial rate
avg dt (ms) = 5.00 +- 2.7047
sample rate (ms)
time (s)
0
2
4
6
8
10
12
10
20
30
40
50
60

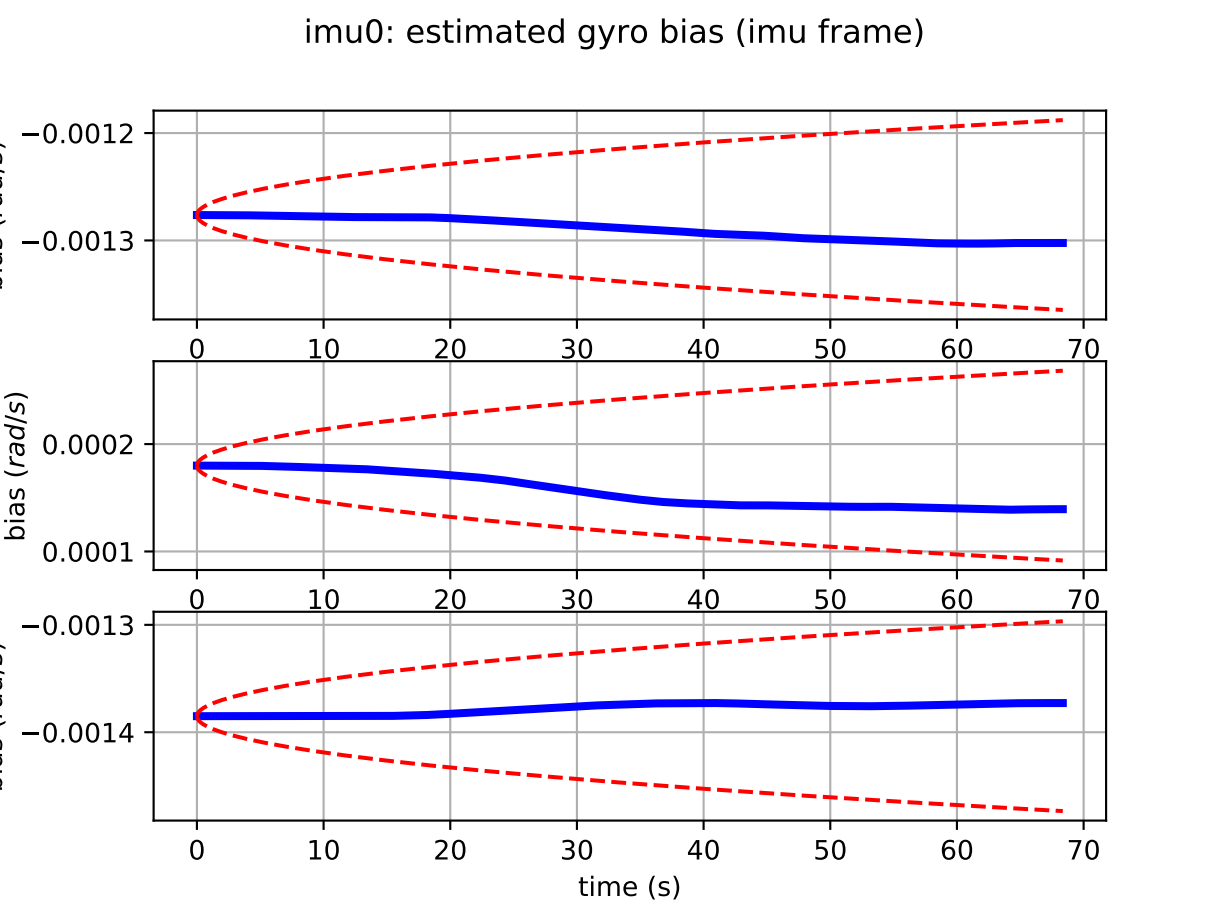

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
2.5
0.0
−2.5
10
8
0
10
20
30
40
50
60
70
time (s)

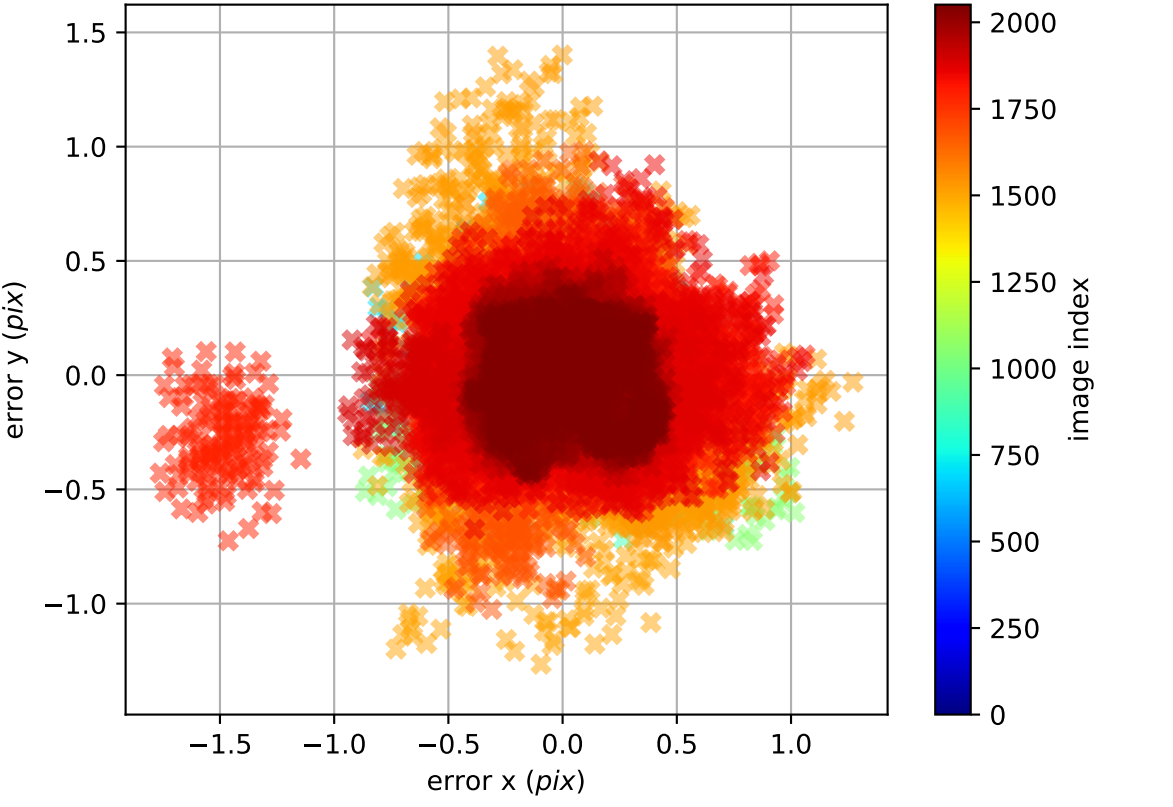

imu0: acceleration error
error (m/s^2)
0.1
0.0
−0.1
0
2000
4000
6000
8000
10000
12000
error index


imu0: estimated accelerometer bias (imu frame)
−0.0028
−0.0030
−0.0038
−0.0040
−0.0028
−0.0030
0
10
20
30
40
50
60
70
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
1
0
−1
0
10
20
30
40
50
60
70
ang. velocity (rad/s)
est. bodyspline
imu0
0.5
0.0
−0.5
0
10
20
30
40
50
60
70
ang. velocity (rad/s)
est. bodyspline
imu0
1
0
−1
0
10
20
30
40
50
60
70
time (s)


imu0: angular velocities error
error (rad/s)
0.02
0.00
−0.02
0
2000
4000
6000
8000
10000
12000
error (rad/s)
0.00
−0.02
0
2000
4000
6000
8000
10000
12000
error (rad/s)
0.01
0.00
−0.01
0
2000
4000
6000
8000
10000
12000
error index


imu0: estimated gyro bias (imu frame)
−0.0012
−0.0013
0.0002
0.0001
−0.0013
−0.0014
bias (rad/s)
0
10
20
30
40
50
60
70
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
2000
1750
1500
1250
1000
750
500
250
0